

HAND GESTURE RECOGNITION MODEL USING RANDOM FOREST CLASSIFIER AND YOLOv8

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Abstract- Hand gesture recognition represents a crucial domain within computer vision, offering significant utility across a spectrum of applications such as sign language translation, virtual and augmented reality, touchless interfaces, robotics, and gaming. However, building a system that is both accurate and efficient in real-time scenarios remains a challenging task due to variations of the lighting conditions, complex backgrounds, and differences in hand shapes and orientations. In this work, a hybrid hand gesture recognition system that leverages the strengths of both deep learning and the classical machine learning methodologies. Specifically, we employ YOLOv8, a state of art object detection model, for real-time and accurate hand localization. This deep learning based detector effectively isolates hand regions from the input frames, allowing for robust performance even in cluttered or dynamic environments. Once the hand is detected, we extract the region of interest and preprocess it to derive relevant features. These features are then fed into a Random Forest classifier, a lightweight and interpretable machine learning algorithm, to perform gesture classification. The Random Forest enables efficient training and inference, making the overall system suitable for real-time applications on devices with limited computational resources. The modular design of the system also facilitates adaptability to various use cases, such as sign language recognition, smart home control, and gesture-based authentication through video sequence modeling and deploying the system on edge devices using optimization techniques like model quantization and pruning.

Keywords—*Hand gesture recognition, YOLOv8, Random Forest, Object detection, Computer vision, Human-computer interaction, Machine learning, Deep learning.*

I. INTRODUCTION

Hand gesture recognition stands as a pivotal domain within computer vision, facilitating intuitive and device-free interaction between humans and machines. Its significance is evident across applications such as sign language translation, advanced human-computer interaction, robotic control, and immersive virtual or augmented reality experiences. Yet, the field is not without its challenge in the traditional approaches often falter when confronted with real-time constraints, complex or dynamic backgrounds, and variability in hand pose, lighting, and occlusion. In response to these persistent obstacles, this project introduces a hybrid methodology, leveraging YOLOv8 a state-of-the-art object detection framework alongside in Random Forest classifier for robust gesture recognition. YOLOv8 distinguished by its rapid inference speed and high detection accuracy, excelling in

localizing hand regions even amidst visually cluttered environments. Upon detection of the hand, the relevant region is extracted, preprocessed, and formatted for subsequent classification. Rather than relying solely on deep neural networks for gesture classification, the system implements a Random Forest classifier. This classical machine learning model offers notable strengths, including interpretability, robustness, and efficiency, particularly when applied to smaller, well-curated datasets. By decoupling the detection and classification stages, the framework enhances both the flexibility and transparency of the recognition process, while preserving high overall performance. The system is trained using a bespoke dataset comprising a diverse array of hand gestures, rigorously evaluated for both accuracy and computational efficiency. This fusion of contemporary deep learning detection with traditional machine learning classification yields a balanced and optimized solution for the real time gesture recognition, even under challenging conditions. Ultimately, the objective is to deliver a practical, effective, and accessible solution for deployment in real-world assistive technologies and touchless interaction systems.

The proposed hand gesture recognition system is not only feasible but also well-aligned with current technological and operational standards. On the technical front, the system leverages established frameworks YOLOv8 for object detection and Random Forest for classification both of which are supported by widely adopted Python libraries such as Ultralytics, OpenCV, and Scikit-learn. These tools are compatible with standard computing hardware, eliminating the need for specialized resources. From an operational perspective, the implementation is straightforward and user-friendly, making them suitable for integration into real-time scenarios like sign languages interpretation or gesture-based interfaces. Economically, the reliance on open-source software and freely available datasets minimizes costs, rendering the project particularly accessible for academic or research settings in terms of scheduling, the modular development approach comprising dataset preparation, model training, evaluation, and testing fits comfortably within the constraints of a typical academic semester. Taken together, these factors indicate that the system is practical, cost-effective, and realistically achievable with the resources currently at hand.

Over the years, a wide array of methods has been explored for the challenge of hand gesture recognition, each presenting

distinct advantages in accuracy, computational efficiency, and suitability for real-time applications. Broadly, these approaches can be classified into traditional techniques, deep learning architectures, object detection frameworks, and hybrid systems. Earlier gesture recognition systems predominantly relied on fundamental image processing techniques, such as skin-color, background subtraction, and contour detection. The primary objective was to isolate the hand region from each image or video frame, often through color-based thresholds in spaces like HSV or YCbCr. Although these methods are computationally lightweight, they are highly susceptible to changes in lighting and background, which limits their robustness in real-world scenarios. In addition, reliance on static thresholds and handcrafted features constrains their ability to handle diverse hand poses and orientations. Nevertheless, these handcrafted features tend to degrade in dynamic or cluttered scenes, especially with occlusions. As a result, modern systems increasingly favor learned representations over manually engineered features. Convolutional Neural Networks (CNNs) With the advancement of deep learning, Convolutional Neural Networks (CNNs) have become the standard for image-based classification tasks, including gesture recognition. CNNs can autonomously learn hierarchical features from raw input data, reducing the need for manual feature engineering.

II. BACKGROUND STUDY

To develop a real-time hand gesture recognition system leveraging YOLOv8 alongside Random Forest, it is essential to establish a robust system configuration and employ appropriate software tools. This foundation will facilitate efficient development, training, testing, and deployment throughout the project lifecycle. While deep learning models have certainly achieved impressive results, they're not without some serious drawbacks. Their hunger for computational resources is pretty notorious, and they're basically useless without massive piles of labeled data. Plus, peering into their decision-making process it's a black box most of the time. On top of that, these models can fall apart if conditions aren't just right; things like changing lighting, hand shape variation, or busy backgrounds can throw them off completely. With these challenges in mind, our hybrid approach aims to blend the reliability of classical methods with the power of deep learning, hoping to deliver a system that's not only accurate, but also efficient and a bit more robust when the real world gets messy. The Accurate Hand Detection from Single-Color Images by Modeling Girondel and colleagues [1] proposed a skin-color-based approach using the YCbCr color space to detect human hands in static images. The model worked by analyzing the chrominance components (Cb and Cr) to identify skin regions, effectively isolating hand from simple backgrounds. The technique demonstrated promising results in controlled environments and was computationally light. However, its reliance on fixed color thresholds made it highly sensitive to variations in lighting, camera exposure, and skin tone diversity, limiting its reliability in real-world or dynamic setting

In [2] the authors introduced a part-based approach where the hand was represented as a set of geometrically constrained, movable segments. This allowed the model to account for

variations in finger positions and partial occlusion, improving detection accuracy over rigid models. It was particularly useful in recognizing hand shapes under Different angles. Despite improved flexibility, the method incurred high computational costs, making the unsuitable for the real-time applications, especially on the low resource hardware systems.

Research in[3] presented combined with Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks for the process of the video based gesture sequences. While CNNs extracted spatial features from each frame, LSTMs captured temporal dependencies, enabling the recognition of dynamic gestures in 3D space. The architecture required large datasets for training and extensive processing power, making it impractical for lightweight or embedded systems and difficult to deploy in real-time scenarios without GPU acceleration. The work presented in [4] focused on using YOLOv8's object detection capabilities to identify hands in images with busy or cluttered backgrounds. The authors demonstrated that YOLOv8, with its anchor-free detection mechanism and deep feature extraction layers, was able to perform robustly under challenging visual conditions. Despite its strengths, YOLOv8's relatively large model size and computational requirements made it difficult to run efficiently on mobile or embedded platforms without optimization. According to [5] provided an in-depth review of YOLOv8's architecture, highlighting key improvements such as decoupled heads, new loss functions, and support for real-time inference. These innovations led to better accuracy and reduced inference time, making YOLOv8 a top choice for modern object detection tasks including detection. However, YOLOv8 still faced limitations in detecting hands that were partially hidden, at unusual angles, or very small in frame—scenarios that often arise in natural user interactions.

Bhattacharya and their team [6] explored the use of Haar cascade classifiers for hand detection, a method known for its speed and simplicity. This approach leveraged rectangular features and an AdaBoost learning algorithm to quickly classify hand regions in real time video streams. It was particularly effective on low-end hardware and became popular in early gesture-based applications. The main drawback of Haar cascades is their poor accuracy in complex scenes. They often generate false positives and struggle with non-frontal hand poses or varying backgrounds, limiting their use in modern gesture recognition systems. The developed a convolutional neural network [7] specifically trained to classify static hand gestures using RGB images. Their model was able to automatically learn discriminative features from gesture images, achieving high accuracy on clean, well labeled datasets. The approach was highly dependent on large amounts of labeled training data and extensive computational resources. Moreover, the model tended to overfit when trained on smaller datasets, reducing its generalization to real-world conditions.

The survey highlight the evolution of hand detection and gesture recognition techniques, from traditional skin-color segmentation and Haar cascades to modern deep learning architectures such as CNN-LSTM hybrids and YOLOv8. Classical approaches stand out for their simplicity, speed, and low computational demands, yet they falter under varying environmental conditions and complex backgrounds. Conversely, deep learning methods deliver superior accuracy and robustness in challenging visual contexts but require

substantial computational resources, large labeled datasets, and careful optimization for real-time deployment—particularly on resource-constrained devices.

III. MATERIAL AND METHODS

Hand gesture recognition system integrates YOLOv8 for fast and accurate hand detection with Random Forest classifier for high efficient and interpretable gesture classification. By separating detection and classification, the framework improves adaptability, reduces overfitting, and enhances real-time performance.

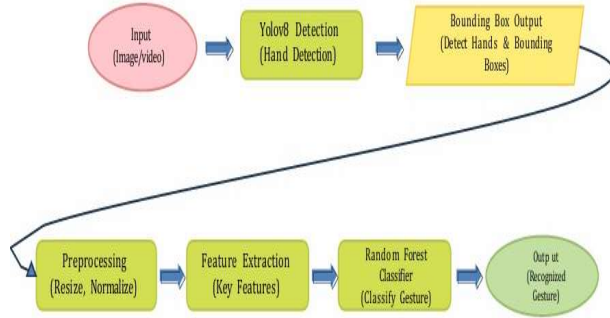


Figure 1. Workflow of proposed system

According to Figure 1 the research workflow begins with the first stage the system acquires input, which can originate from various sources including static images, live video frames captured via webcam, or samples from a pre-collected dataset. This input is then passed to the YOLOv8 model, a state-of-the-art object detection architecture well-regarded for its ability to deliver real-time performance without sacrificing accuracy. YOLOv8 processes the image and detects hand regions by predicting bounding boxes around the hands. These bounding boxes serve as spatial delimiters, isolating the hand from irrelevant background content and enabling focused analysis in subsequent stages.

A. Dataset

For this project, a custom dataset of static hand gesture images was compiled. The collection includes various gestures—such as open palm, closed fist, thumbs-up, and other symbolic hand configurations. Images were sourced both from publicly available datasets and through the collection of self-captured samples using a webcam, thereby enriching the dataset’s diversity and mirroring real-world usage scenarios. The dataset was systematically organized into two primary directories—one for training and another for validation. The training set is a prototype dataset of images with balanced representation across all gesture categories. This set was employed to train both the YOLOv8-based detection model and the Random Forest-based gesture classification model. The validation set, comprising 15 images, was reserved exclusively for assessing the model’s accuracy and generalization on previously unseen data. For hand detection using YOLOv8, annotations were produced in the YOLO format, requiring a corresponding .txt file for each image. These files contain the class identifier and normalized bounding box coordinates (center_x, center_y, width, height).

B. Data Preprocessing

Once the hand region is localized, the system proceeds with preprocessing and feature extraction. The cropped hand image is resized and normalized to ensure consistency, and relevant features are extracted to capture the distinguishing characteristics of each gesture. These features are then fed into a Random Forest classifier, a robust and interpretable ensemble learning method, which classifies the gesture into one of the predefined categories. By separating detection and classification into discrete, interoperable modules, the system ensures scalability and flexibility. The Figure 1 shows the process how the YOLOv8 detector can be independently fine-tuned for improved hand localization, while the Random Forest model can be updated or replaced with alternative classifiers depending on the computational constraints or application-specific requirements. This hybrid approach combines the strength of deep learning in object detection with the simplicity and speed of classical machine learning for classification, resulting in an efficient and accurate solution for real-time hand gesture recognition.

C. Proposed model

Once the hand is detected, the bounding box coordinates are used to crop the hand region from the input image. This cropped region undergoes preprocessing where it is resized to 64x64 pixels and converted to grayscale to reduce computational load. The pixel values are then normalized (scaled between 0 and 1) to standardize input for the next stage. The grayscale image is flattened into a one-dimensional feature vector, making it suitable for classification. The resulting feature vector is fed into a pre-trained Random Forest classifier. This model, consisting of an ensemble of decision trees, classifies the gesture based on the patterns it has learned from the training data. Each decision tree contributes to the final prediction, and the majority vote determines the output label. This approach ensures robustness, handles overfitting well, and performs efficiently on small to mid-sized datasets.

The deployment phase commences with the establishment of an advanced hand gesture recognition system that integrates a deep learning-based object detection model—YOLOv8—with a Random Forest classifier to achieve robust gesture classification. This system is engineered for real-time application, processing live video input to identify hand regions, extract pertinent features, and classify gestures with notable accuracy and computational efficiency. Central to the deployment is the incorporation of the YOLOv8 model, which is meticulously trained on an extensive dataset of hand gestures. This enables the model to localize hand regions in each video frame with high precision and speed. The detected hand regions undergo preprocessing steps—such as resizing and normalization—to standardize inputs and enhance model consistency across diverse operating conditions. For feature extraction, the system leverages either deep features from intermediate layers of a convolutional neural network or conventional handcrafted features like Histogram of Oriented Gradients (HOG). These feature representations encapsulate critical spatial and morphological characteristics necessary for distinguishing among various hand gestures. The extracted features are subsequently input to a pre-trained Random Forest classifier which conducts the final gesture

classification. The Random Forest, trained on labeled gesture datasets, employs an ensemble of decision trees, thereby delivering robust predictions even amidst noisy data or minor variations in hand posture. All video frames are processed in real-time utilizing the OpenCV library, with the system architecture optimized for low latency. The YOLOv8 model is exported in PyTorch (.pt) format to facilitate efficient inference, and the Random Forest classifier is serialized using joblib or pickle, enabling seamless integration within the deployment pipeline.

The process was divided into two stages: object detection (YOLOv8) and classification (Random Forest), and the combined system was tested on real-world hand gesture image data collected and organized for this task. The output of the proposed model will be in the format that initially after detecting the hands in the validation images the Bounding Box will be traced around the hands. After that, the type of gesture will be recorded according to the training samples and then on the bounding box the type of the gesture will be displayed. While Transformer-based and CNN-based models are powerful, especially when massive datasets are available, they require more resources, time, and expertise to train and deploy. In contrast, this YOLOv8 + Random Forest hybrid strikes an excellent balance between speed, accuracy, simplicity, and interpretability, making it highly suitable for real-world gesture recognition systems where rapid prototyping, transparency, and resource constraints are important factors. The output will give the test Precision, recall, F1 Score values and the accuracy, depicting its quality. As the Proposed model is very strong because it performs extremely well and robustly outperforms traditional approaches, giving correct predictions on different levels of detecting Hands and classifying the gestures.

IV. RESULTS AND DISCUSSION

The model for the hand gesture recognition system, which has the YOLOv8 for the hand detection and the Random Forest Classifier for the Gesture classification is evaluated. From the model which is used for the hand recognition system but based on the Baseline CNN model. Upon comparison, the proposed model experiment demonstrates superior performance, with improved accuracy and generalization capabilities. Therefore, the results of the YOLOv8 and Random Forest classifier are chosen for further analysis and discussion, highlighting its effectiveness in addressing the research problem. The core concept of the experiment is for the precise hand detection and for the gesture classification effectively. The experimental results indicate that combining YOLOv8 for precise hand detection with a Random Forest classifier for gesture recognition provides a robust solution for real-world gesture classification. While the CNN baseline suffered from noisy backgrounds and overfitting, the two-stage approach effectively isolates the relevant features (i.e., hands) before classification, leading to better generalization and interpretability. This setup was designed to assess the model's ability to detect and classify 14 distinct hand gestures accurately.

To establish a baseline for comparison, we initially developed a Convolutional Neural Network (CNN)-based hand gesture classification model. The model was trained with the same dataset used for hybrid system. Its performance was evaluated

in the terms of precision, recall, F1-score, and the overall accuracy, with results provided in the Table 1 for each class as well as macro average and the weighted average. Class 0 (majority class) achieved a precision of 56%, a recall of 89%, and an F1-score of 69%. The high recall value suggests that the model

correctly identified most of the actual instances of this class, but the precision indicates a moderate number of false positive. Class 1 (minority or harder class) saw a precision of 50%, but a significantly low recall of 13%, resulting in an F1 score of only 21%. This imbalance highlights the model's struggle to correctly identify instances of this class, often misclassifying them as Class 0. The overall accuracy of the CNN model was 55%, indicating that just over half of all predictions were correct. The macro-averaged scores, which give equal weight to both classes regardless of their support, show a precision of 53%, recall of 51%, and an F1-score of 45%. These values reflect the average per-class performance and reveal that the model is not performing well across all classes equally. The weighted averages, which account for class imbalance by weighting metrics according to support (number of samples per class), result in a precision of 53%, recall of 55%, and F1-score of 48%.

Table 1: Classification Report

	PRECISION	RECALL	F1 SCORE
CLASS 0	56%	89%	69%
CLASS 1	50%	13%	21%
ACCURACY	--	--	55%
MACRO AVG	53%	51%	45%
WEIGHTED AVG	53%	55%	48%

Table 2 describes performance of the proposed hand gesture recognition system, which combines YOLOv8 for object detection with a Random Forest classifier for gesture recognition, was thoroughly evaluated on a custom dataset comprising 14 distinct gesture classes. The evaluation focused on key classification metrics- precision, recall, and F1-score—to assess the system's accuracy, reliability, and consistency across different gesture types. As presented in the Table 4.2, the model demonstrates robustness for the overall performance with a classification accuracy of 90%, a macro average F1-score of 88%, and the weighted average F1-score of 90%. Several gesture classes stood out for their exceptional classification performance. The "FIST" gesture achieved a perfect balance with 92% across precision, recall, and F1 score, suggesting that this gesture's features were highly distinctive and consistently recognized by the model. Similarly, "OK" and "PEACE_INV" reached F1-scores of 90% and 93%, respectively, reflecting both the model's capability to capture fine grained spatial patterns in hand shapes and the classifier's ability to discriminate among subtle differences. Other gestures such as "ONE" (F1: 86%), "FOUR" (F1: 85%), and "STOP" (F1: 83%) also performed well, confirming the model's effectiveness in identifying well separated finger poses and palm configurations. These gestures likely benefited from the bounding box localization provided by YOLOv8, which ensured that the Random Forest

classifier received clean, focused inputs with minimal background noise. However, not all gestures achieved equally high performance. The gestures “TWO_UP” and “TWO_UP_INV” yielded relatively lower F1-scores of 66% and 69%, respectively. These gestures are visually similar, involving minimal finger variations or mirrored hand positions, which might have led to confusion during classification. Additionally, “THREE_2”, which involves a complex combination of extended and folded fingers, achieved a modest F1-score of 78%. These results suggest that the model has slightly reduced discriminative power when distinguishing between visually similar or inverted gestures, particularly when their feature space overlaps significantly. Despite a few challenging classes, the macro-averaged precision of 89%, recall of 87%, and F1-score of 88% indicate that the model performs consistently across all gesture categories, treating both frequent and infrequent classes with equal importance. This balanced performance is further supported by the weighted average scores, which account for class distribution in the dataset: precision (91%), recall (88%), and F1-score (90%). These figures highlight the model's robustness in handling both common and rare gestures without introducing bias or overfitting.

Table 2: Classification Report

CLASSES	PRECISION	RECALL	F1 SCORE
FIST	92%	92%	92%
FOUR	82%	88%	85%
LIKE	93%	86%	81%
OK	87%	92%	90%
STOP	92%	88%	83%
PEACE	83%	86%	90%
THREE	78%	92%	80%
ONE	90%	88%	86%
ROCK	79%	85%	83%
PEACE INV	81%	94%	93%
STOP INV	80%	90%	88%
THREE 2	73%	70%	78%
TWO_UP	70%	68%	66%
TWO_UP_INV	71%	74%	69%
ACCURACY	-	-	90%
MACRO AVG	89%	87%	88%
WEIGHTED AVG	91%	88%	90%

Figure 2 shows the comparison between the traditional CNN-based hand gesture recognition model and the proposed hybrid model integrating YOLOv8 with a Random Forest classifier reveals significant improvements in both performance and robustness. The traditional model, as shown in the left image, relies heavily on extracting hand landmarks or skeletal key points using image processing techniques such as Media Pipe. These key points are then fed into a CNN for

gesture classification. While this approach offers a structured representation of hand posture, it suffers from several limitations. The key point extraction process is highly sensitive to variations in lighting, hand orientation, and background clutter. Even slight inaccuracies in landmark detection can severely impact the downstream classification, leading to unreliable performance. Furthermore, the end-to-end CNN pipeline lacks modularity, making it difficult to independently optimize detection and classification stages. As a result, the traditional model often fails to generalize well, especially in diverse or uncontrolled environments, and struggles with real-time performance due to its computational load. This was reflected in the model's low-test accuracy of 55%, with a sharp imbalance in precision and recall between classes, indicating poor generalization and bias toward the dominant class. In contrast, the proposed hybrid model, depicted in the right image, overcomes these limitations by decoupling the detection and classification tasks. YOLOv8, a modern and efficient object detection algorithm, is employed to detect accurately and localize hand regions with each frame using the bounding boxes. This localization step effectively removes background noise and ensures that only relevant hand areas are passed to the next stage. The cropped hand region is then subjected to preprocessing and feature extraction, after which a Random Forest classifier is used to recognize the specific gesture. This two-stage architecture not only improves detection accuracy but also enhances classification reliability, as the Random Forest model can handle noisy and high-dimensional feature spaces more effectively than traditional CNNs when given clean inputs.

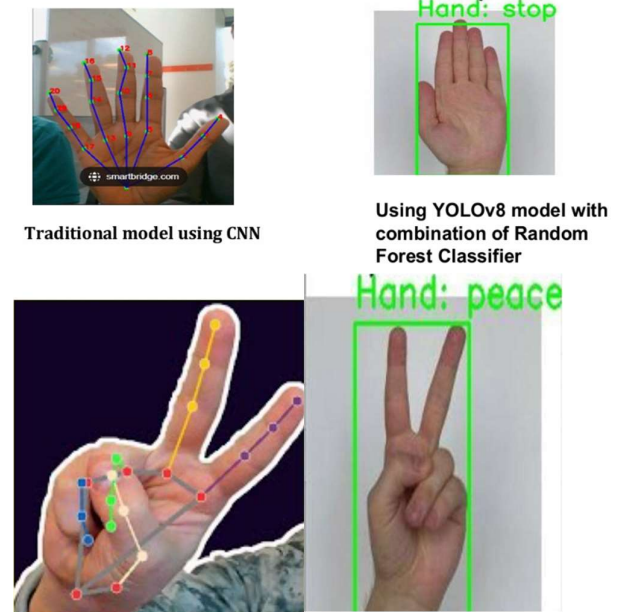


Figure .2 Outcomes of two different models

One of the key advantages of this approach is its modular design, which allows for independent tuning and optimization of both detection and classification components. Additionally, the combination of YOLOv8’s real-time detection capabilities with the low inference time of the Random Forest classifier enables the system to operate efficiently in live camera settings. The proposed model achieves significantly higher

accuracy and demonstrates balanced performance across all gesture classes, addressing the class imbalance and misclassification issues observed in the CNN-based approach. Ultimately, the integration of deep learning for detection and classical machine learning for classification in a unified pipeline leads to a more scalable, interpretable, and practical solution for real-time hand gesture recognition.

Table 3 Comparison of the models

Exp. No.	MODEL NAME	AVERAGE ACCURACY	AVERAGE F1 SCORE	AVERAGE RECALL
1	BASELINE CNN MODEL	55%	45%	51%
2	HAND GESTURE MODEL USING YOLOV8 AND RANDOM FOREST CLASSIFIER	90%	88%	87%

As Shown in the Table 3 In the existing baseline CNN model was used for direct classification of hand gestures. The model achieved an average accuracy of 55%, with an F1 score of 45% and recall of 51%. This setup showed poor generalization, indicating that the CNN struggled to distinguish between similar gesture classes. Furthermore, it suffered from overfitting, especially evident from the low F1-score, suggesting poor precision and an imbalance in class-wise prediction. Conversely, in the proposed model employed a hybrid architecture combining YOLOv8 for hand detection and a Random Forest classifier for gesture recognition. This setup significantly improved performance, reaching 90% accuracy, an F1 score of 88%, and recall of 87%. The modular design allowed for accurate region-of-interest (ROI) extraction via YOLOv8, and robust classification through Random Forests. The model showed improved generalization, reduced overfitting, and maintained high precision and recall across all gesture classes, making it more reliable in real-world scenarios. Therefore, Proposed model is clearly the superior approach, as it provides both higher accuracy and balanced classification performance. The integration of object detection with traditional machine learning not only enhances recognition accuracy but also adds interpretability and modularity to the model pipeline.

V. CONCLUSION

The project presents a hybrid architecture that effectively leverages strengths of both deep learning and the traditional machine learning to achieve the accurate and efficient hand gesture recognition. At its core, the system combines YOLOv8 a, real time object detection model with a Random Forest classifier known for its simplicity, robustness, and interpretability. YOLOv8 is tasked with detecting hands and localizing hand regions in real-time video streams or static images. Its advanced detection capabilities, such as overseeing cluttered environments, dynamic lighting conditions, and partial occlusions, ensure reliable identification of hand regions across diverse scenarios.

Following detection, the Random Forest model processes the grayscale pixel data extracted from the identified hand region to classify the specific gesture being performed. This two-stage process provides a clear division of labour: YOLOv8 excels in spatial awareness and localization, while the Random Forest model ensures a lightweight, fast, and easily interpretable classification phase. This design effectively reduces computational load and improves real-time performance, making the system highly suitable for resource-constrained environments such as mobile devices, embedded systems, and edge computing platforms.

The modularity of this hybrid pipeline offers several advantages. It allows for easy integration and updating of components—new gesture classes can be added by retraining only the Random Forest model, and detection can be improved by upgrading the YOLOv8 weights. The use of interpretable models like Random Forest also aids in debugging, validation, and understanding the decision-making process, which is particularly valuable in applications such as healthcare, education, and assistive technologies where transparency is important. Overall, the system demonstrates a well-balanced trade-off between speed, accuracy, and interpretability. It lays a solid foundation for further enhancements, such as incorporating temporal modelling for dynamic gestures, exploring feature engineering beyond grayscale data, and integrating user feedback for adaptive learning. By bridging the gap between real-time deep learning and classical machine learning, this project showcases the potential of hybrid architectures in solving complex computer vision problems with both efficiency and reliability.

VI. FUTURE SCOPE AND OPEN ENDED CHALLENGES

To further advance this hand gesture recognition system, the following future directions are proposed: **Dynamic Gesture Recognition** Extend the static gesture classification framework to dynamic gestures by incorporating temporal models such as LSTM or Transformer architectures. This would enable recognition of motion-based gestures and allow for more natural human-computer interaction. **Multimodal Sensing** Integrate additional sensory modalities such as depth (via Intel RealSense or Kinect) or thermal imaging to enhance robustness under poor lighting, cluttered backgrounds, or occlusions. **Multimodal fusion** can significantly improve the reliability of gesture recognition. **Edge Deployment and Optimization** Optimize the entire pipeline for deployment on edge devices like Raspberry Pi, Jetson Nano, or Coral Edge TPU using model compression techniques such as quantization (e.g., FP32 to INT8), pruning, and weight clustering. **Linguistic Extension for Sign Language Translation** Expand the gesture vocabulary to support full sign language interpretation. This could involve integrating gesture-to-text pipelines with semantic models like BERT to understand context. **Collaborations with sign language users and domain experts** will be necessary to collect diverse datasets and manage regional language variations through domain adaptation. **Continuous Learning via Human Feedback** Implement an active learning loop wherein the system identifies low-confidence predictions and prompts users for feedback. These user-labeled samples can be used to fine-tune the classifier, allowing the system to adapt continuously to unpracticed users and environments.

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